

WEIL POSITIVITY ON THE WINDOW OF WIDTH $\log 10$: AN INTERVAL-ARITHMETIC CERTIFICATE

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ABSTRACT. Weil’s criterion states that the Riemann Hypothesis holds if and only if Weil’s quadratic form Q is nonnegative on all admissible test functions. Restricting the support of the test functions to a fixed interval gives a weaker, unconditional question, and positivity is known for small intervals. We prove that $Q(f) \geq 2^{-49162} \|f\|_2^2$ for every complex C^2 function f supported in $[-\frac{1}{2} \log 10, \frac{1}{2} \log 10]$. The half-width $\frac{1}{2} \log 10 \approx 1.1513$ exceeds that of the previous proofs we know of: $\frac{1}{2} \log 5$ (formal proof in Lean), 0.8 (Zhu), and $17/16$ (Liu, unrefereed). The proof follows Liu’s bounded-comparison framework. The prime-power shifts, now including $n = 9$, are kept inside a bounded operator on the whole space, and the high frequencies are controlled by a lower bound on the out-of-band mass. This reduces positivity to a finite matrix inequality of size 320 in each parity. We certify that inequality with all quadrature, rounding and inversion errors paid: the finite margin is about 8.45×10^{-44} (even) and 4.40×10^{-40} (odd), while the total error is below 10^{-55} . The certificate is produced and checked by separate programs in Arb ball arithmetic and was reimplemented independently. It is an external certificate, not a formal proof. The tail estimate is proved under the condition $L < 7/6$; larger windows require new estimates and a new certificate. The result concerns one fixed window and does not address the Riemann Hypothesis.

Version 2 (10 October 2026), Zenodo doi:10.5281/zenodo.23282866; version 1: doi:10.5281/zenodo.23238480. Version 2 removes from the status box the statement about AI reviews; nothing else changes. Certificate data and code: release v1.4.0 of <https://github.com/moriryota/prolate-weil-bounds>, doi:10.5281/zenodo.23092422. Licence: CC BY 4.0.

Status. Analytic proof plus an external interval-arithmetic certificate (Arb/FLINT). The certificate was produced and checked by separate programs, and the final check was reimplemented independently. Not formalized in Lean. Not yet reviewed by independent human experts.

1. INTRODUCTION

1.1. Background. Let Q be Weil’s quadratic form on test functions f on the additive line $\mathbb{R}$ (the logarithm of the multiplicative variable) [10, 2]. Weil’s criterion states that the Riemann Hypothesis (RH) is equivalent to $Q(f) \geq 0$ for all admissible f . If the support of f is restricted to an interval $[-L, L]$, the statement becomes a question about finitely many primes, since only prime powers $n < e^{2L}$ contribute, and it can be decided unconditionally for small L . Yoshida proved positivity for $L = \frac{1}{2} \log 2$ by an analytic reduction combined with a computer-assisted check of a 200×200 matrix [11]. Connes and Consani gave a different, archimedean approach, comparing the archimedean distribution with a Sonin trace for test functions whose Fourier transforms vanish at $\frac{i}{2}$ and 0 [3]; this is not the same statement for arbitrary test functions. Zhu [12] certified positivity for $L = 0.8$ by interval arithmetic. The author proved it for $L = \frac{1}{2} \log 5$ in Lean 4, inside the Lean kernel [7]. Liu [6] claims a certificate for $L = 17/16$ with an explicit lower bound $2^{-49162} \|f\|_2^2$.

For larger windows, with $\mu = e^{2L} \geq 50$, the bottom of the spectrum of the restricted form is at most $C\mu^{9/2} \log \mu e^{-4\pi\mu}$ [9], and numerical computations suggest that it is of this size. If so, a certificate has to resolve a positive margin of this size, which needs a precision of order e^{2L} bits, together with a rigorous control of everything outside the finite computation.

1.2. Result. Let $L = \frac{1}{2} \log 10$ and $I = [-L, L]$. For $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset I$ put $F_f(z) = \int f(x) e^{izx} dx$,

$$a(t) = \text{Re } \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi,$$

and

$$(1.1) \quad \begin{aligned} Q(f) = & 2 \left| \int f(x) \cosh \frac{x}{2} dx \right|^2 - 2 \left| \int f(x) \sinh \frac{x}{2} dx \right|^2 + \frac{1}{2\pi} \int_{\mathbb{R}} a(t) |F_f(t)|^2 dt \\ & - 2 \sum_{n \in \{2, 3, 4, 5, 7, 8, 9\}} \frac{\Lambda(n)}{\sqrt{n}} \text{Re} \int_{\mathbb{R}} f(x) \overline{f(x - \log n)} dx. \end{aligned}$$

By the Guinand–Weil explicit formula, $Q(f) = \text{Re} \sum_{\rho} m_{\rho} F_f(\gamma_{\rho}) \overline{F_f(\overline{\gamma_{\rho}})}$, the sum over the distinct nontrivial zeros ρ with multiplicities m_{ρ} and $\gamma_{\rho} = (\rho - \frac{1}{2})/i$. RH is not assumed (Section 2).

Theorem L. For every $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset [-\frac{1}{2} \log 10, \frac{1}{2} \log 10]$,

$$Q(f) \geq 2^{-49162} \int_{\mathbb{R}} |f(x)|^2 dx.$$

In particular $Q(f) > 0$ for $f \not\equiv 0$.

The proof is an ordinary analytic argument together with an external certificate in ball arithmetic (Arb/FLINT [1]), produced and checked by separate programs. It is not a formal proof.

reference	half-width L	e^{2L}	method
Yoshida [11]	$\frac{1}{2} \log 2 \approx 0.347$	2	analytic + computer-assisted
Zhu [12]	0.8	4.95	interval certificate
[7]	$\frac{1}{2} \log 5 \approx 0.805$	5	Lean 4 (kernel-checked)
Liu [6]	$17/16 = 1.0625$	8.37	interval certificate (unrefereed)
Theorem L	$\frac{1}{2} \log 10 \approx 1.151$	10	interval certificate

TABLE 1. Unconditional positivity on windows $[-L, L]$. The approach of Connes and Consani [3] works under vanishing conditions on the Fourier transform and is not listed.

1.3. Method and what is new. We follow the framework of Liu [6], which we credit for the structure of the argument:

- (1) The prime-power shifts are kept inside a bounded operator $M = \frac{7}{2}I - C_p$ on the whole space $H = L^2(I)$, and the frequencies $|t| > \Omega = 256$ are controlled by a lower bound on the out-of-band mass. This gives $Q(f) \geq \tau \|f\|^2 + \langle f, (M + V)f \rangle$ with a bounded V (Section 3).
- (2) A block elimination reduces $M + V \succeq 0$ on H to a finite matrix inequality on the first $N = 640$ Legendre modes, with the infinite remainder paid by explicit projection errors (Section 4).

New in this paper:

- the window $L = \frac{1}{2} \log 10$, where the prime power 9 enters, with the operator bound $\|C_p\| \leq 3$ for all seven prime powers;
- a self-contained proof of the tail inequality, including the out-of-band bound, which does not assume Liu’s results;
- an optimized upper bound for the inverse compression $E^* M^{-1} E$ (Section 4). With it the finite condition holds at $N = 640$; for $N \leq 256$ the corresponding matrix had negative smallest eigenvalue;

- analytic bounds for the quadrature errors of all source integrals, including the digamma symbol on complex ellipses (Section 5);
- a certificate with separate proposer and checker, mutation tests, and an independent reimplement of the final check (Section 6).

The constant 2^{-49162} is the same as Liu's, because it comes from the same out-of-band bound at the same Ω . The finite part has a much larger margin, about 10^{-43} . The out-of-band estimate is proved here under the convenient condition $L < 7/6$ (that is, $e^{2L} < 10.3$); larger windows require new estimates, not only a larger computation.

2. SETTING

Throughout, $L = \frac{1}{2} \log 10$, $I = [-L, L]$, $H = L^2(I; \mathbb{C})$ with Lebesgue measure, and $F_f(t) = \int_I f(x) e^{itx} dx$, so that $\frac{1}{2\pi} \int |F_f|^2 = \|f\|^2$. The weights are $\Lambda(p^k) = \log p$, and only $n < e^{2L} = 10$ contribute: $n \in \{2, 3, 4, 5, 7, 8, 9\}$.

Explicit formula. Let $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset I$. Then $f = f' = 0$ at $\pm L$. Two integrations by parts give $F_f(z) = O((1 + |\text{Re } z|)^{-2})$ uniformly in $|\text{Im } z| \leq \frac{1}{2}$. Together with $N(T) = O(T \log T)$, the zero sum converges absolutely. The autocorrelation $h(x) = \int f(v+x) \overline{f(v)} dv$ is compactly supported, $h(-x) = \overline{h(x)}$, it is at least C^4 , and its transform is $F_f(z) \overline{F_f(\bar{z})}$. The Guinand–Weil explicit formula applied to the real even part of h gives (1.1). The poles contribute $2(|C|^2 - |S|^2)$ with $C = \int f \cosh \frac{x}{2}$ and $S = \int f \sinh \frac{x}{2}$, since $F_f(\pm \frac{i}{2}) = C \mp S$. The primes contribute $-2 \sum \Lambda(n) n^{-1/2} \text{Re } h(\log n)$, and the gamma factor gives the term with $a(t)$. The symmetries $\rho \mapsto 1 - \rho$ and $\rho \mapsto 1 - \bar{\rho}$ show that the zero sum is real. No assumption on the location of the zeros is made.

Parities and complex functions. Q is additive over real and imaginary parts. All operators below are real, self-adjoint and commute with the reflection $x \mapsto -x$, so it suffices to treat the even and odd subspaces separately. Positivity of real symmetric matrices holds in complex coordinates.

3. THE BOUNDED COMPARISON

3.1. The prime operator. For $a \in \mathbb{R}$ let $S_a f(x) = \mathbf{1}_I(x-a) f(x-a)$, and put

$$C_p = \sum_{n \in \{2,3,4,5,7,8,9\}} \frac{\Lambda(n)}{\sqrt{n}} (S_{\log n} + S_{-\log n}), \quad M = \frac{7}{2} I - C_p.$$

Then $-2 \sum \Lambda(n) n^{-1/2} \text{Re} \int f \overline{f(\cdot - \log n)} = -\langle f, C_p f \rangle$.

Lemma 3.1. $\|C_p\| \leq 3$. Hence $\frac{1}{2} I \preceq M \preceq \frac{13}{2} I$.

Proof. For a positive weight w on I , the weighted Schur (Young) inequality $2|f(x)f(x-\ell)| \leq |f(x)|^2 w(x-\ell)/w(x) + |f(x-\ell)|^2 w(x)/w(x-\ell)$ gives $|\langle f, C_p f \rangle| \leq \int_I R_w |f|^2$ with

$$R_w(x) = \sum_n \frac{\Lambda(n)}{\sqrt{n}} \frac{\mathbf{1}_I(x + \log n) w(x + \log n) + \mathbf{1}_I(x - \log n) w(x - \log n)}{w(x)}.$$

With $w(x) = 1 + \frac{19}{40}(x/L)^2$, R_w is even. On $x \geq 0$ the set of active shifts changes only at $q = e^{2x} \in \{1, \frac{10}{9}, \frac{8}{5}, \frac{5}{2}, \frac{49}{10}, \frac{32}{5}, \frac{81}{10}, 10\}$. Bounding R_w on 1792 open subintervals in ball arithmetic gives $R_w < 3$ almost everywhere, that is, outside the finitely many breakpoints (at the closed endpoint $q = \frac{5}{2}$ the value is about 3.22, which does not matter). $\square$

3.2. The tail. Let $\Omega = 256$, let B_{band} be the operator on H with kernel $\sin(\Omega(x-y))/(\pi(x-y))$ (the compression of the band projection $|t| \leq \Omega$), and

$$K = 2|\cosh \frac{x}{2}\rangle\langle\cosh \frac{x}{2}| - 2|\sinh \frac{x}{2}\rangle\langle\sinh \frac{x}{2}| + \frac{1}{2\pi} \int_{-\Omega}^{\Omega} (a(t) - \frac{7}{2})|e_t\rangle\langle e_t| dt, \quad e_t(x) = e^{itx}.$$

Since $\frac{1}{2\pi} \int \frac{7}{2}|F_f|^2 = \frac{7}{2}\|f\|^2$, for admissible f ,

$$(3.1) \quad Q(f) = \langle f, (M + K)f \rangle + T_{\text{tail}}(f), \quad T_{\text{tail}}(f) = \frac{1}{2\pi} \int_{|t| > \Omega} (a(t) - \frac{7}{2})|F_f(t)|^2 dt.$$

Lemma 3.2 (out-of-band mass). $I - B_{\text{band}} \succeq \delta I$ on H , with $\delta = 2^{-49158}$.

Proof. Let $\|f\| = 1$ and $E = \frac{1}{2\pi} \int_{|t| > 256} |F|^2$, with $F = F_f$. If $E \geq \frac{1}{2}$ we are done. Otherwise the in-band integral exceeds π , so some $t_0 \in [-256, 256]$ has $|F(t_0)| > \sqrt{\pi/512} > \frac{1}{16}$. By Cauchy–Schwarz, $|F^{(j)}(t)| \leq \sqrt{2L}L^j < 2L^j$ for all real t and $j \geq 0$, since $L < 7/6$.

Let $N = 4096$, $h = \frac{1}{32}$ and $I_j = [256 + 2jh, 256 + (2j+1)h]$ for $0 \leq j < N$. On each I_j some x_j has $|F(x_j)|^2 \leq 2\pi E/h < 256E$. Put $G(t) = \text{Re}(e^{-i \arg F(t_0)} F(t))$, a real function, and interpolate G at the nodes x_j . The nodes satisfy $|x_j - x_k| \geq h|j - k|$ and $|t_0 - x_j| < 768$. The Lagrange basis therefore satisfies

$$\sum_j |\ell_j(t_0)| \leq \left(\frac{768}{h}\right)^{N-1} \sum_{j=0}^{N-1} \frac{1}{j!(N-1-j)!} = \frac{49152^{4095}}{4095!} < 2^{24570}.$$

The interpolation remainder is

$$|R(t_0)| \leq \frac{2(768L)^{4096}}{4096!} \leq 2\left(\frac{3 \cdot 768L}{4096}\right)^{4096} < 2(2/3)^{4096} < 2^{-2047}.$$

Hence $\frac{1}{16} < 16\sqrt{E} \cdot 2^{24570} + \frac{1}{32}$, so $E > 2^{-49158}$. $\square$

The hypothesis $L < 7/6$ is used in the remainder bound $(3 \cdot 768L/4096 < 2/3)$. It is a convenient condition for this argument, not a necessary limit; for larger L the parameters, and the other estimates below, would have to be redone.

Lemma 3.3 (digamma). For $|t| \geq 256$, $a(t) - \frac{7}{2} > \frac{123}{1280} =: C_0$.

Proof. For $t > 0$ let $w = \frac{1}{4} + \frac{it}{2}$ and $z = w + 1$. Binet's formula [4, 5.9.15] and the recurrence [4, 5.5.2] give

$$\psi(w) = \log z - \frac{1}{2z} - \frac{1}{w} - 2 \int_0^\infty \frac{u du}{(u^2 + z^2)(e^{2\pi u} - 1)}.$$

Here $|u^2 + z^2| \geq |\text{Im } z^2| = \frac{5t}{4}$, $\int_0^\infty u(e^{2\pi u} - 1)^{-1} du = \frac{1}{24}$, $\text{Re } \log z \geq \log \frac{t}{2}$, $\text{Re } \frac{1}{2z} \leq \frac{5}{2t^2}$ and $\text{Re } \frac{1}{w} \leq \frac{1}{t^2}$. Hence $a(t) \geq \log \frac{t}{2\pi} - \frac{1}{t}$ for $t \geq \frac{15}{4}$, and a is even. Finally $\log \frac{256}{2\pi} > \frac{18}{5}$, by $\pi < \frac{22}{7}$, $e < \frac{68}{25}$ and $(448/11)^5 > (68/25)^{18}$. $\square$

Proposition 3.4 (tail inequality). Let $S = I - B_{\text{band}} - \delta I$, $v_0 = (2L)^{-1/2}$, $v_1 = \sqrt{3/(2L)} x/L$, $h_j = S v_j$, and $U = 81|h_0\rangle\langle h_0| + 27|h_1\rangle\langle h_1|$. Then for admissible f , with $\tau = 2^{-49162}$,

$$T_{\text{tail}}(f) \geq \tau\|f\|^2 + \langle f, Uf \rangle.$$

Proof. By Lemma 3.3, $T_{\text{tail}}(f) \geq C_0\langle f, (I - B_{\text{band}})f \rangle = C_0\langle f, Sf \rangle + C_0\delta\|f\|^2$, and $C_0\delta > \delta/16 = \tau$. Next, $0 \preceq S \preceq I$ by Lemma 3.2. With $c = 256L \in [272, \frac{896}{3}]$, direct integration gives

$$e_0 := \langle v_0, (I - B_{\text{band}})v_0 \rangle = \frac{2}{\pi} \int_c^\infty \frac{\sin^2 x}{x^2} dx, \quad e_1 = \frac{6}{\pi} \int_c^\infty \frac{(\cos x - \sin x/x)^2}{x^2} dx.$$

Integration by parts gives

$$\frac{c-1}{\pi c^2} \leq e_0 \leq \frac{c+1}{\pi c^2}, \quad \frac{3(c-2)}{\pi c^2} \leq e_1 \leq \frac{3c^2+6c+2}{\pi c^3}.$$

Hence $81e_0 < C_0$ and $27e_1 < C_0$, while $e_0, e_1 > 2^{-11} > \delta$. By parity $\langle v_0, Sv_1 \rangle = 0$. Minimizing $\langle f - z_0 v_0 - z_1 v_1, S(f - z_0 v_0 - z_1 v_1) \rangle \geq 0$ over $z_0, z_1 \in \mathbb{C}$ gives

$$\langle f, Sf \rangle \geq \frac{|\langle h_0, f \rangle|^2}{e_0 - \delta} + \frac{|\langle h_1, f \rangle|^2}{e_1 - \delta},$$

and $C_0/(e_j - \delta)$ exceeds 81 and 27. $\square$

By (3.1) and Proposition 3.4, with $V = K + U$ (a bounded self-adjoint operator),

$$(3.2) \quad Q(f) \geq \tau \|f\|^2 + \langle f, (M + V)f \rangle.$$

It therefore suffices to prove $M + V \succeq 0$ on H . Note that U is built from Sv_j , not from v_j : projecting onto v_j itself would add an artificial positive term in the almost-null low-frequency directions.

4. REDUCTION TO A FINITE MATRIX

Let $v_j(x) = \sqrt{(2j+1)/(2L)} P_j(x/L)$, $E : \mathbb{C}^N \rightarrow H$ with $Ec = \sum_{j < N} c_j v_j$, $P = EE^*$, $N = 640$ (even and odd together; 320 in each parity), and

$$A = E^*ME, \quad B = E^*M^2E, \quad J = E^*VE, \quad G = E^*M^{-1}E, \quad D_M = B - A^2 \succeq 0.$$

B is the compression of M^2 , not A^2 .

Block elimination [6]. Write $R = V - EJE^*$, so that $PRP = 0$. Suppose $\|(I - P)RP\| \leq e$, $(I - P)R(I - P) \succeq -n$ and $\|(I - P)R(I - P)\| \leq h$ with $n < m = \frac{1}{2}$. For $\theta, \chi > 0$ put

$$\alpha = e\theta + \frac{(1 + \chi)e^2}{m - n}, \quad \beta = \frac{e/\theta + n}{m^2} + \frac{(1 + \chi^{-1})h^2}{m^2(m - n)}.$$

Eliminating the complementary block of M (with $L_c = -M_c^{-1}F$ for the cross block F , so that $L_c^*L_c \preceq m^{-2}D_M$) and applying Young's inequality to the three blocks of R shows

$$(4.1) \quad G^{-1} + J \succeq \alpha I + \beta D_M \implies M + V \succeq 0 \text{ on } H.$$

Projection errors. The Chebyshev ellipse $\rho = \frac{3}{2}$ gives

$$\sup_{|t| \leq 256} \|(I - P)e_t\|^2 \leq 72Le^{5\Omega L/6}(2/3)^{2N} < 2^{-380}$$

(an integer comparison), so one may take $r = 2^{-190}$. The band part of K has weight $|a(t) - \frac{7}{2}|$ with $-7 < a(t) < 7$ on $|t| \leq 256$, so its absolute weight integral is $\kappa = \frac{1}{2\pi} \int_{-256}^{256} |a(t) - \frac{7}{2}| dt < 896$. Moreover $\|h_0\| < \frac{1}{29}$, $\|h_1\| < \frac{3}{50}$ and $\|(I - P)h_j\| < 10r$, and the pole projection error is below $p = 2^{-440}$ (it is at most $48/640!$). Then $e \leq \kappa\sqrt{2L}r + 10(\frac{81}{29} + \frac{81}{50})r + p$, $n \leq \kappa r^2 + p$ and $h \leq (\kappa + 10800)r^2 + p$, so one may take

$$e = 1500r + p, \quad n = 896r^2 + p, \quad h = 12000r^2 + p,$$

and with $\theta = \chi = 1$ one checks $n < m$ and $\alpha + \beta(13/2)^2 < 2^{-170}$.

Inverse upper bound. G^{-1} is not computable directly. For any matrix X and $0 < \mu_0 \leq m$ put

$$\mathcal{W}(X) = X + X^* - X^*AX + \mu_0^{-1}(I - AX - X^*A + X^*BX).$$

With $R_X = E - MEX$, we have $\mathcal{W}(X) - G = R_X^*(\mu_0^{-1}I - M^{-1})R_X \succeq 0$, hence $\mathcal{W}(X)^{-1} \preceq G^{-1}$. With $\mu_0 = m$ and $C = B - mA \succ 0$, completing the square shows that the best choice is $X_* = C^{-1}(A - mI)$:

$$\mathcal{W}(X) - \mathcal{W}(X_*) = m^{-1}(X - X_*)^*C(X - X_*) \succeq 0, \quad \mathcal{W}(X_*) = m^{-1}\{I - (A - mI)C^{-1}(A - mI)\}.$$

So the condition to certify, in each parity, is

$$(4.2) \quad T = \mathcal{W}^{-1} + J - \alpha I - \beta(B - A^2) \succ 0.$$

Remark 4.1. The simple compression $A + J$ of $M + V$ is positive already for small N , but its positivity does not imply $M + V \succeq 0$ on H , because the complementary block is ignored. The valid condition uses G^{-1} , or its lower bound $\mathcal{W}^{-1}$. In our computations the midpoint value of the smallest eigenvalue of $\mathcal{W}^{-1} + J$ was negative for $N = 128, \dots, 256$ (for example -9.9×10^{-25} in the even parity at $N = 256$), and positive at $N = 640$. In addition, at $N \leq 256$ the projection error does not allow $n < m$.

5. THE SOURCE MATRICES AND THEIR QUADRATURE ERRORS

All matrices are computed in Arb ball arithmetic [1]. Ball arithmetic encloses rounding errors only, so the quadrature errors are bounded separately, as operator norms in the orthonormal coordinates.

A and B. After splitting I at the support boundaries of the shifts, the entries of A and B are integrals of polynomials of degree at most 1278. Gauss–Legendre quadrature with 642 nodes is exact up to degree 1283, so there is no truncation error. The declared errors $\epsilon_A = \epsilon_B = 2^{-400}$ cover the rounding and the distance to the rational centres; this is checked for every entry.

J: the band integral. With the spherical Bessel functions j_j ,

$$\int_I v_j(x) e^{itx} dx = \sqrt{2L(2j+1)} i^j j_j(Lt).$$

In each parity this gives real vectors $b_j(t)$, and

$$[B_{\text{band}}]_{ij} = \frac{1}{\pi} \int_0^{256} b_i b_j dt, \quad [K_{\text{band}}]_{ij} = \frac{1}{\pi} \int_0^{256} (a(t) - \frac{7}{2}) b_i b_j dt.$$

These are computed on 10 panels $[0, \frac{1}{2}]$, $[\frac{1}{2}, 1]$, $\dots$, $[128, 256]$ with 192 Gauss nodes each.

Lemma 5.1. *On the Bernstein ellipses $\rho = 2$ of the panels, $|b_i b_j| \leq 2Le^{96L} < 3 \cdot 10^{48}$. The function $g(z) = \frac{1}{2}[\psi(\frac{1}{4} + \frac{iz}{2}) + \psi(\frac{1}{4} - \frac{iz}{2})] - \log \pi - \frac{7}{2}$ is holomorphic with $|g| < 1000$. Consequently the band and gamma-weighted quadrature errors are at most $320 \cdot 1024 \cdot 3 \cdot 10^{48} \cdot 2^{-383}$ and $320 \cdot 1024 \cdot 4000 \cdot 10^{48} \cdot 2^{-383}$ in operator norm, for all 640 modes at once.*

Proof. $F_j(z) = \int_I v_j e^{izx} dx$ is entire, $|F_j(z)| \leq \sqrt{2L} e^{L|\text{Im } z|}$, and the representation by j_j has a removable singularity at 0. The enlarged ellipses lie in $|\text{Im } z| \leq 48$, $|z| \leq 272$. On them $w = \frac{1}{4} \pm \frac{iz}{2}$ satisfies $|w + k| \geq \frac{1}{16}$ for all integers $k \geq 0$, so all poles of ψ are excluded. Shifting 25 times to $v = w + 25$ with $\text{Re } v \geq \frac{5}{4}$, the series [4, 5.7.6] gives $|\psi(v)| < 325$, and the recurrence costs at most $25 \cdot 16$. A Chebyshev truncation of degree 383 on $\rho = 2$ and the exactness of 192-point Gauss up to degree 383, with positive weights summing to the panel length, give the panel error $4\ell H 2^{-383}$. The sum of the panel lengths is 256, and the operator norm is at most 320 times the entrywise bound. $\square$

Poles and U. The coefficients of the cosh and sinh vectors are computed with 680 Gauss nodes. Their quadrature error is bounded by comparison with the Taylor polynomials of degree 720 (for which 680-point Gauss is exact against v_j , $j < 640$), whose remainder is $d_p = 2(3/5)^{721}/721!$; the two rank-one terms then cost at most $160d_p$. For U , $\|E^* S v_j\| \leq 1$, so the column error ϵ_b costs $81(2\epsilon_b + \epsilon_b^2)$. Altogether

$$\epsilon_{J, \text{quad}} < 7.47 \times 10^{-59} < 2^{-190},$$

and the declared $\epsilon_J = 2^{-185}$ also covers the rounding.

6. THE CERTIFICATE

Rational data. For each parity the proposer outputs dyadic rational matrices with 512 binary digits: centres A_0, B_0, J_0 , a matrix X , $W_0 \approx \mathcal{W}(X)$, $Y \approx W_0^{-1}$, and three congruence transforms for C , W and T . The floating-point and Cholesky computations of the proposer are not trusted.

Error propagation. With $x \geq \|X\|$, $a \geq \|A_0\|$ and $b = \frac{13}{2}$, the checker verifies

$$(x^2 + 4x)\epsilon_A + 2x^2\epsilon_B + \|W_0 - \mathcal{W}(A_0, B_0; X)\| \leq \epsilon_W = 2^{-350}, \quad \epsilon_B + 2a\epsilon_A + \epsilon_A^2 \leq \epsilon_D = 2^{-380}.$$

Then $\widehat{W} = W_0 + \epsilon_W I \succeq \mathcal{W}(X) \succeq G \succeq b^{-1}I$. For $R = I - \widehat{W}Y$ with $\|R\| \leq r_Y < 1$, $\|\widehat{W}^{-1} - Y\| \leq \|Y\|r_Y/(1 - r_Y) =: \epsilon_Y$, and $\|\widehat{W}^{-1} - \mathcal{W}(X)^{-1}\| \leq 2b^2\epsilon_W$. Hence

$$T \succeq T_0 - \epsilon_T I, \quad T_0 = Y + J_0 - \alpha I - \beta(B_0 - A_0^2), \quad \epsilon_T = 2b^2\epsilon_W + \epsilon_Y + \epsilon_J + \beta\epsilon_D.$$

For a rational Z with $\|Z^*T_0Z - I\| \leq \eta$ and $\|Z\|_F^2 \leq s$, if $\eta + \epsilon_T s < 1$ then $\lambda_{\min}(T) \geq (1 - \eta - \epsilon_T s)/s$. Also $\mathcal{W}(X_*) \preceq \mathcal{W}(X)$, so the certified bound also covers the optimal $\mathcal{W}$. The checker also verifies $C \succ 0$ and $\widehat{W} \succ 0$.

	even	odd
total error ϵ_T	2.04×10^{-56}	2.04×10^{-56}
certified lower bound for $\lambda_{\min}(T)$	$8.4536864638 \times 10^{-44}$	$4.4015025569 \times 10^{-40}$
compared with $2^{-144} = 4.48 \times 10^{-44}$	pass	pass

TABLE 2. The certified finite condition (4.2) at $N = 640$.

Checks. The checker does not import the proposer and computes no eigenvalues. It compares every entry of the rational centres with the saved source balls (whose SHA-256 sums are pinned), and it reports a specific reason for each rejection. In mutation tests it rejected all five altered inputs for the intended reasons: the sign of J reversed; an inflated declared error; B replaced by A^2 ; a file changed without updating its hash; and a negative T . An independent reimplementaion at 1536 bits, with a more conservative norm bound for nonsymmetric residuals, reproduced both lower bounds to 24 digits. The checker runs in about 20 seconds and 1 GB of memory.

Proof of Theorem L. The certificate gives (4.2), hence $G^{-1} + J \succeq \alpha I + \beta D_M$ in each parity. By (4.1), $M + V \succeq 0$ on H . By (3.2), $Q(f) \geq 2^{-49162}\|f\|^2$ for every admissible f . $\square$

7. REMARKS

Margins. The compression of Q to the first 192 Legendre modes has smallest eigenvalue about 1.54×10^{-43} in the even sector and 7.46×10^{-40} in the odd sector (numerical, not certified; cf. the Galerkin computations of [9]). The certified finite margins 8.45×10^{-44} and 4.40×10^{-40} are of the same order. The final constant 2^{-49162} is much smaller only because of the crude out-of-band bound in Lemma 3.2. The best possible constant in Lemma 3.2 is $\inf_{\|f\|=1} \frac{1}{2\pi} \int_{|t|>\Omega} |F_f(t)|^2 dt = 1 - \lambda_0(\Omega L)$, the defect of the top eigenvalue of time–frequency limiting. Inserting Fuchs’ asymptotic formula $4\sqrt{\pi}\sqrt{c}e^{-2c}$ with $c = \Omega L$ gives about 2^{-844} [5, 8]; this is an asymptotic estimate, not a proved bound.

Limits of the method. Lemma 3.2 is proved here for $L < 7/6$. A rigorous lower bound for $1 - \lambda_0(c)$ valid for all c would remove this condition and improve τ . Larger windows also need more prime powers in M , possibly a larger Ω , larger N and more precision, since the margin to be resolved is expected to be of the size discussed in the introduction.

Trust. The proof relies on the analytic lemmas above, on Arb/FLINT [1] for the ball enclosures (Gauss nodes and weights, logarithms, Bessel and digamma values), and on the checker program. It has not been checked by a proof assistant, unlike [7].

REPRODUCIBILITY

The source matrices, the certificate, the proposer, the checker, the mutation tests, the independent reimplementations of the final check and the scripts for the analytic and scalar checks are in the folder `certificate-VI/` of release `v1.4.0` of <https://github.com/moriryota/prolate-weil-bounds> (about 225 MB, with SHA-256 sums and replay instructions), archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions). The checker runs in about 20 seconds; regenerating the source matrices takes about 6 minutes.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude and OpenAI GPT. Under the direction of the author, these systems designed the reduction, derived the analytic bounds, implemented and ran the computations, and reviewed one another's work adversarially. The review included an independent reimplementations of the final check. The author is responsible for the content.

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